University of Michigan

NERS 590

Advanced Reactor Design

OpenMC Neutronics Analysis of an ABR-1000 Reactor Core

Authors:

Cal Kaufman

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Abstract

A continuous-energy Monte Carlo neutronics and depletion model of the Advanced Burner Reactor (ABR-1000) sodium-cooled fast reactor (SFR) was developed using OpenMC. The model represents an equilibrium transuranic (TRU)-recycled metal fuel core based on the ANL-AFCI-202 reference design [1] and the OECD/NEA benchmark specifications [5]. A homogenized cylindrical approximation was employed to represent the 379-assembly hexagonal core, with five-zone axial and two-zone radial fuel composition stratification consistent with benchmark definitions. The depletable fuel inventory is discretized into ten distinct material regions (five axial, two radial), and a predictor-corrector (CECM) integration scheme is used over a 12-step geometric timestep schedule spanning 328.5 effective full-power days (EFPD).

The final model achieves a beginning-of-cycle (BOC) effective multiplication factor of $k_{\text{eff}} \approx 1.04916$, in reasonable agreement with the benchmark average of 1.0355 ± 0.0078 [5]. Depletion analysis reproduces key fuel cycle performance metrics, including a burnup reactivity swing of approximately -2415 pcm (benchmark: 2210 ± 422 pcm) and a TRU consumption rate of approximately 94.8 kg/year (ANL reference: 81.6 kg/year).

Remaining discrepancies relative to deterministic benchmark participants are attributed primarily to geometric simplifications inherent in the cylindrical homogenization and the use of natural molybdenum as a pseudo-fission product surrogate in lieu of an explicit fission product chain. The model provides a physically consistent framework for further studies of fast reactor fuel cycle performance and equilibrium core behavior.

Contents

1 Introduction **3**

2 Core Design Description **3**
2.1 Reference Core Layout . . . 3
2.2 Axial Configuration . . . 5
2.3 Fuel Composition . . . 5
2.4 Control Assemblies . . . 6

3 Neutronics Modeling and Methodology **6**
3.1 Geometry Approximation . . . 6
3.2 Material Homogenization . . . 8
3.3 Nuclear Data . . . 8
3.4 Depletion Methodology . . . 8
3.5 Transport Settings . . . 9
3.6 Tallies and Post-Processing . . . 9

4 Results and Analysis **9**
4.1 Eigenvalue Performance . . . 9
4.2 Depletion Behavior . . . 11
4.3 Burnup Reactivity Swing . . . 12
4.4 Transuranic Consumption . . . 13
4.5 Actinide Inventory Evolution . . . 14
4.6 Power Distribution . . . 16
4.7 Neutron Spectrum and Leakage . . . 17

5 Discussion **18**
5.1 Sources of Reactivity Bias . . . 18
5.2 Depletion Accuracy . . . 19
5.3 Geometric Simplification Effects . . . 19
5.4 Limitations and Future Work . . . 20

6 Conclusion **20**

1 Introduction

Sodium-cooled fast reactors (SFRs) occupy a central role in advanced nuclear fuel cycle strategies owing to their ability to efficiently fission transuranic (TRU) isotopes recovered from light water reactor (LWR) spent nuclear fuel. By operating on a hard neutron spectrum, SFRs achieve TRU conversion ratios substantially below unity, enabling sustained actinide burning that reduces the long-term radiotoxicity and heat load of nuclear waste [1]. The Advanced Burner Reactor (ABR-1000), developed by Argonne National Laboratory (ANL) under the Global Nuclear Energy Partnership (GNEP) program, is a 1000 MWt pool-type SFR designed for a closed fuel cycle with recycled metallic U-TRU-Zr fuel and a target TRU conversion ratio of approximately 0.7 [1].

Accurate neutronics modeling of SFR cores presents several challenges that distinguish them from thermal-spectrum systems. The fast spectrum demands high-fidelity cross-section data across the unresolved resonance region, where probability table sampling is essential. The absence of significant thermalization also reduces the sensitivity of k_{eff} to geometric approximations, though the spatial distribution of fissile material and the treatment of leakage remain important. In depletion calculations, the complex multi-isotope TRU composition evolves rapidly under high fast fluences, requiring consistent tracking of minor actinides and accurate transmutation chains.

Monte Carlo methods provide a rigorous framework for capturing these effects without reliance on group-collapsing approximations, diffusion theory, or spatial homogenization of individual pins. The OpenMC code [4] offers continuous-energy transport, a modern depletion interface via the openmc.deplete module, and direct compatibility with ENDF/B-VIII.0 cross-section libraries, making it well-suited to this class of problems.

In this work, OpenMC is used to develop a continuous-energy model of the ABR-1000 equilibrium TRU-recycled metal core. Eigenvalue and full-cycle depletion calculations are performed to evaluate reactivity, power distribution, and fuel cycle performance. The model is measured against published ANL and OECD/NEA results [1–3, 5], with particular attention to sources of bias arising from the cylindrical geometry approximation and the homogenized material treatment adopted in this work.

2 Core Design Description

2.1 Reference Core Layout

The ABR-1000 reference core is a compact, pool-type SFR of 1000 MWt rated thermal power, designed for a one-year irradiation cycle (328.5 EFPD at 90% capacity factor) with a four-batch fuel management scheme [1]. The hexagonal core lattice employs an assembly pitch of 16.142 cm with a flat-to-flat duct dimension of 15.710 cm and an inter-assembly gap of 0.432 cm. The full core contains 379 assemblies: 180 driver fuel assemblies divided into 78 inner-core and 102 outer-core positions, 114 radial reflector assemblies, 66 radial shield

assemblies, and 19 control assemblies (15 primary and 4 secondary) [1,5]. Figure 1 illustrates the reference radial core layout.

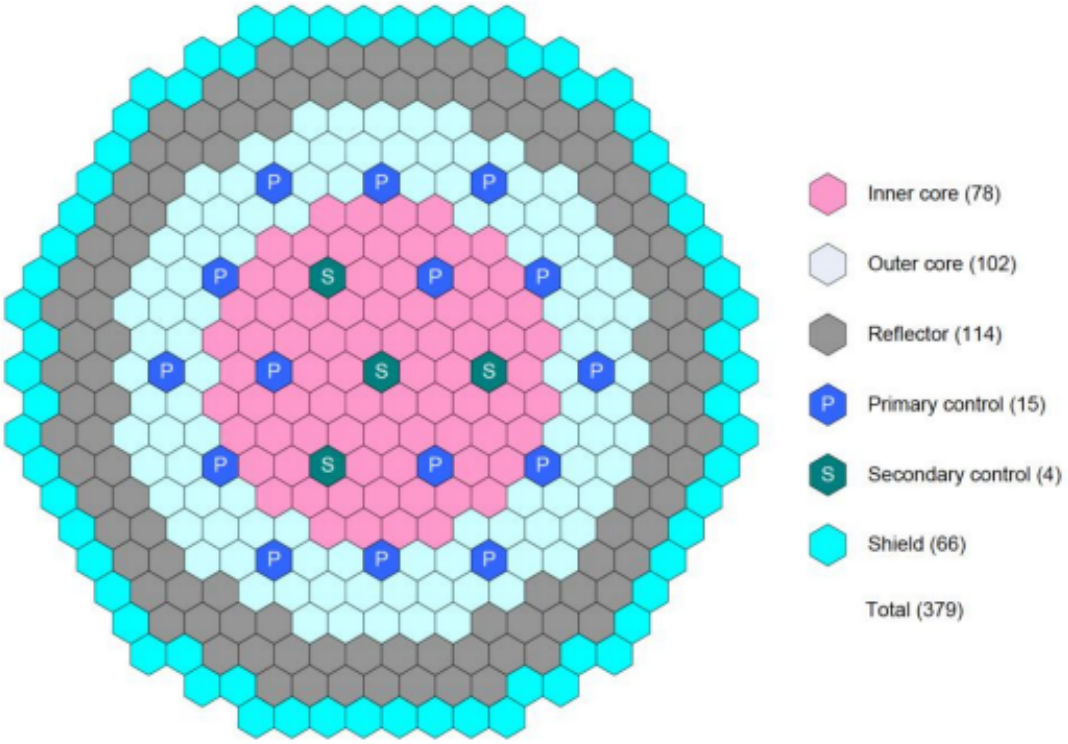

Figure 1: Radial layout of the 1000 MWt ABR reference core showing inner core (78 assemblies), outer core (102 assemblies), radial reflector (114 assemblies), radial shield (66 assemblies), and primary (P) and secondary (S) control assemblies. Adapted from Ref. [1].

Each fuel assembly contains 271 fuel pins arranged on a triangular pitch with wire-wrap spacers. The fuel pin consists of a ternary metallic alloy (U-TRU-10wt%Zr) bonded with sodium, clad in HT9 ferritic-martensitic stainless steel. The pin pitch-to-diameter ratio is 1.180, the cladding outer diameter is 0.755 cm, and the nominal active core height is 81.3 cm for the metal fuel [1]. Key assembly design parameters are summarized in Table 1.

Table 1: Key fuel assembly design parameters for the ABR-1000 metal fuel core [1, 5].

Parameter	**Value**
Assembly pitch (cm)	16.142
Number of fuel pins per assembly	271
Fuel pin cladding material	HT9
Cladding outer diameter (cm)	0.755
Pin pitch-to-diameter ratio	1.180
Active core height, operating (cm)	85.82
Fuel volume fraction (%)	39.00
Coolant (Na) volume fraction (%)	35.34
Structural (HT9) volume fraction (%)	25.66
Bond material	Sodium
Fuel smeared density (% TD)	75.0

2.2 Axial Configuration

The assembly has a total height of 477.52 cm. The axial stack, from bottom to top, consists of: lower structure (35.76 cm, homogenized SS-316 and sodium), lower axial reflector (125.16 cm, HT9 and sodium), active fuel zone (85.82 cm), displaced bond sodium region (20.06 cm), fission gas plenum (101.01 cm), and upper structure (112.39 cm), for a total active stack height of 480.20 cm [5]. The active core is axially divided into five uniform composition zones in this model to represent the burnup-dependent axial composition gradient at the beginning of equilibrium cycle (BOEC).

2.3 Fuel Composition

The equilibrium TRU-recycled metal fuel composition at BOEC is derived from the OECD/NEA benchmark specification [5]. TRU recovered from ABR spent fuel constitutes the primary feed, supplemented by TRU from LWR spent nuclear fuel (50 MWd/kg burnup, 10-year cooling) as makeup. The resulting fuel contains uranium, plutonium, neptunium, americium, and curium isotopes, with natural molybdenum used as a simplified pseudo-fission product surrogate consistent with benchmark practice [5].

The inner and outer core zones carry distinct TRU vectors and heavy-metal number densities to achieve the target radial power flattening. Representative zone-averaged actinide number densities at BOEC are listed in Table 2.

Table 2: Selected zone-averaged actinide number densities at BOEC (atoms/barn-cm) [5].

Nuclide	**Inner Core (avg.)**	**Outer Core (avg.)**
^{238}U	1.951×10^{-2}	1.820×10^{-2}
^{239}Pu	2.194×10^{-3}	2.789×10^{-3}
^{240}Pu	1.307×10^{-3}	1.724×10^{-3}
^{241}Am	1.073×10^{-4}	1.845×10^{-4}
^{244}Cm	6.591×10^{-5}	7.894×10^{-5}

2.4 Control Assemblies

The primary control system uses boron carbide absorber assemblies with two B-10 enrichment levels: natural boron (19.9 at% ^{10}B) for the fourth-row assemblies and 60% enriched boron for the seventh-row assemblies. Secondary control assemblies use natural boron. In this model, control assemblies are accounted for by volume-fraction blending of the absorber material into the fuel zone, assuming rods fully withdrawn (BOEC condition), consistent with benchmark practice [5].

3 Neutronics Modeling and Methodology

3.1 Geometry Approximation

The hexagonal 379-assembly core is represented using a homogenized cylindrical approximation, wherein each hexagonal ring of assemblies is collapsed into a concentric annular region of equivalent cross-sectional area. The effective radius of an annulus enclosing n assemblies is given by:

$$R_{\text{eff}}(n) = \sqrt{\frac{n \cdot A_{\text{hex}}}{\pi}}, \quad A_{\text{hex}} = \frac{\sqrt{3}}{2}p^2 \tag{1}$$

where $p = 16.142$ cm is the assembly pitch and A_{hex} is the area of a regular hexagon. This yields the following radial boundaries:

$$R_{\text{inner}} = R_{\text{eff}}(85) = 78.13708 \text{ cm} \tag{2}$$
$$R_{\text{outer}} = R_{\text{eff}}(199) = 119.55679 \text{ cm} \tag{3}$$
$$R_{\text{refl}} = R_{\text{eff}}(313) = 149.94082 \text{ cm} \tag{4}$$
$$R_{\text{shield}} = R_{\text{eff}}(379) = 164.99369 \text{ cm} \tag{5}$$

The model employs vacuum boundary conditions on all outer surfaces (radial and axial), consistent with the benchmark specification [5]. Schematics of the homogenized cylindrical model are shown in Figures 2 and 3.

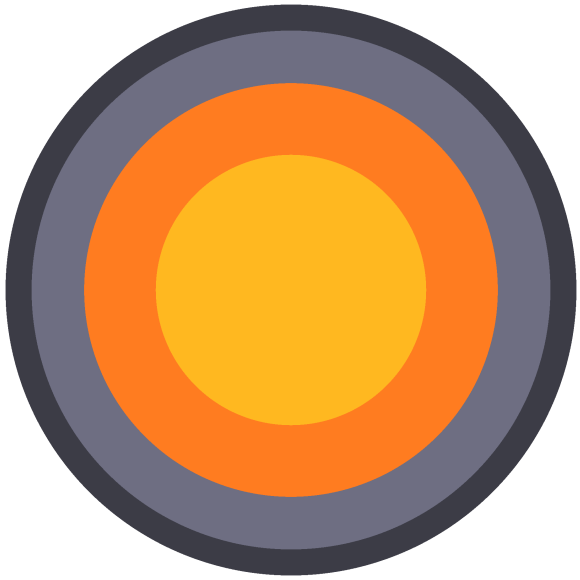

Figure 2: Schematic radial cross-section of the homogenized cylindrical geometry model showing the inner fuel zone (yellow), outer fuel zone (orange), radial reflector (grey), and radial shield regions (black) with vacuum boundary at the outer radius.

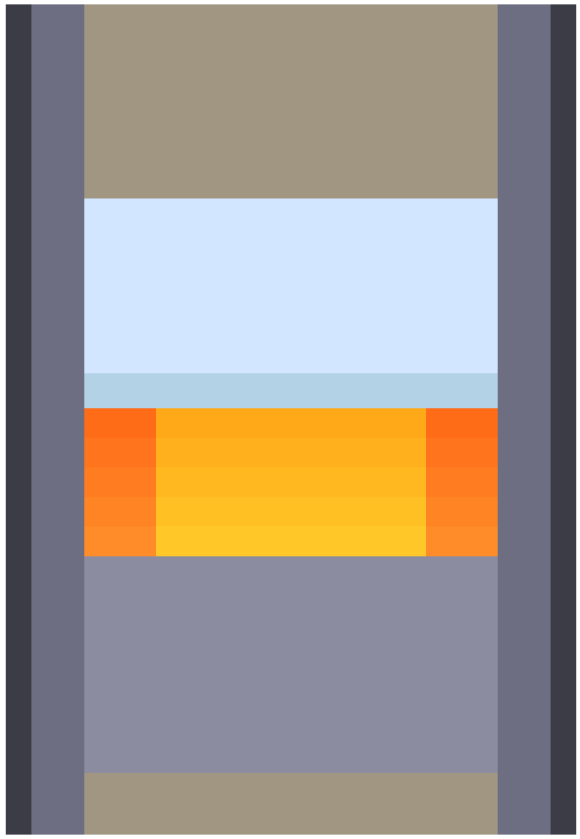

Figure 3: Schematic axial cross-section of the homogenized cylindrical geometry model showing the inner fuel zone (yellows), outer fuel zone (oranges), radial reflector (dark grey), radial shield regions (black), structure (grey), gas plenum (light blue), bond sodium (blue),

and bottom reflector material (dark blue) with vacuum boundary at the outer radius.

3.2 Material Homogenization

Within each radial and axial zone, materials are homogenized at the assembly level. The active-core material in each zone is a volume-fraction-weighted mixture of fuel (39.0%), sodium coolant (35.34%), and HT9 structural material (25.66%) [5]. Number densities for each constituent are taken directly from the benchmark specification (Tables 2.20–2.23 of Ref. [5]) and scaled by the corresponding volume fraction. The control assembly contribution is incorporated by blending the empty-duct (withdrawn rod) material into the active zone at the appropriate assembly fraction: $f_{\mathrm{ci}} = 7/85$ for the inner zone and $f_{\mathrm{co}} = 12/114$ for the outer zone.

Non-fuel axial regions (lower structure, lower reflector, displaced bond sodium, gas plenum, and upper structure) are represented as binary sodium–HT9 mixtures at volume fractions consistent with Tables 2.20 of Ref. [5]. The radial reflector is a dense HT9/sodium mixture (84.5% HT9), and the radial shield contains boron carbide with natural isotopic enrichment in addition to HT9 and sodium [5].

3.3 Nuclear Data

Continuous-energy cross sections from the ENDF/B-VIII.0 nuclear data library are used throughout. Temperature-dependent cross sections are applied using OpenMC's built-in interpolation scheme. Material temperatures are set to $T_{\mathrm{fuel}} = 807.15$ K (average fuel temperature corresponding to 534.0°C) and $T_{\mathrm{struct}} = T_{\mathrm{cool}} = 705.65$ K (coolant and structural average corresponding to 432.5°C), consistent with the nominal operating condition specified in the benchmark [5]. No S(α,β) thermal scattering treatment is applied, as is appropriate for liquid sodium and metallic structural materials at these temperatures.

3.4 Depletion Methodology

Depletion is performed using the openmc.deplete.CoupledOperator interface with the ENDF/B-VIII.0 SFR transmutation chain (chain_endfb80_sfr.xml). The predictor–corrector mid-point (CECM) integration scheme is employed, which requires two transport solves per timestep and achieves second-order accuracy in the burnup evolution [4]. This method is appropriate for the moderate burnup rates characteristic of the ABR cycle (≈73.2 kW/kg HM).

The timestep schedule consists of 12 geometric steps spanning the full 328.5 EFPD cycle, computed via numpy.geomspace between 0.5 d and 328.5 d. Early timesteps (on the order of days) resolve the short-lived fission product equilibration transients, including the ^{135}Xe equilibrium (∼40 h) and the ^{149}Sm equilibrium (∼4 d). Later timesteps (up to ∼138 d) track the slow actinide evolution. The timestep schedule is presented in Table 3.

Table 3: Depletion timestep schedule: 12 geometric steps from 0.5 to 328.5 EFPD. Burnup is computed at the reference power of 1000 MWt and reference heavy-metal inventory of 13,200 kg.

Step	Δt **(d)**	**Cumulative (d)**	**BU (MWd/kg)**
1	0.36	0.4	0.027
2	0.62	1.0	0.074
3	1.06	2.0	0.154
4	1.82	3.8	0.291
5	3.12	7.0	0.527
6	5.35	12.3	0.933
7	9.19	21.5	1.629
8	15.78	37.3	2.824
9	27.10	64.4	4.878
10	46.53	110.9	8.402
11	79.90	190.8	14.455
12	137.69	328.5	24.886

3.5 Transport Settings

Eigenvalue calculations use 60 total batches with 20 inactive batches and 20,000 particles per batch (40 active batches). These settings are appropriate for depletion-coupled transport, where absolute accuracy of k_{eff} is less critical than the accuracy of the one-group transmutation cross sections. The initial fission source distribution is sampled uniformly within the fissile region, with a 20×20×20 Shannon entropy mesh spanning the active core volume to monitor source convergence.

3.6 Tallies and Post-Processing

Eight tallies are defined per transport calculation: total power (kappa-fission, fission, nu-fission), zone-resolved inner and outer power, neutron balance (absorption, nu-fission, (n,2n), (n,3n)), system leakage (current on the outer boundary surfaces), and a 33-group spectral flux tally using the energy group structure from Ref. [5]. Post-processing reads the depletion_results.h5 archive and computes k_{eff} evolution, actinide inventory, burnup reactivity swing, and TRU consumption rate.

4 Results and Analysis

4.1 Eigenvalue Performance

The BOC eigenvalue computed by the model is:

$$k_{\text{eff,BOC}} = 1.04916 \pm 0.00070 \tag{6}$$

(one standard deviation, Monte Carlo statistical uncertainty).

The benchmark average for the MET-1000 core is 1.0355 ± 0.0078, where the spread represents the standard deviation across 20 independent participant calculations using diverse codes and nuclear data libraries [5]. The deviation of the present result from the benchmark mean is approximately 1258 pcm. This level of agreement is consistent with the known spread in Monte Carlo calculations using ENDF/B-VIII.0 relative to JEFF-3.1: the CEA-10 Monte Carlo calculation (TRIPOLI-4, JEFF-3.1, heterogeneous) achieves $k_{\text{eff}} = 1.0429$ [5], while the ANL homogeneous ENDF/B-VII.0 calculation (ANL-4) yields $k_{\text{eff}} = 1.0224$ [5], a spread of approximately 1926 pcm attributable to library differences, geometric modeling, and code bias. The present homogeneous cylindrical model with ENDF/B-VIII.0 is expected to fall within this range, with competing biases: the cylindrical geometry approximation tends to increase leakage and reduce k-eff, while the ENDF/B-VIII.0 library and absence of explicit fission products introduce positive reactivity bias. The net effect is a result slightly above the benchmark mean.

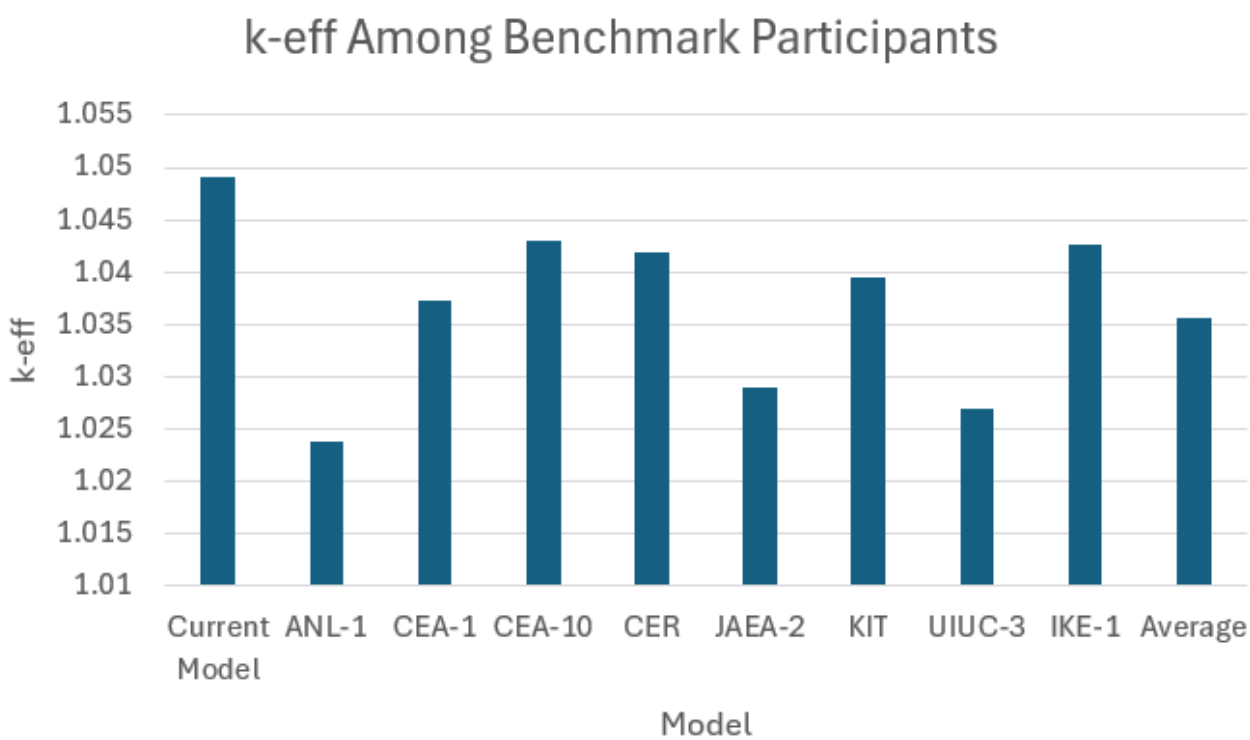

Figure 4: Comparison of computed k_{eff} at BOC with benchmark participant results from Ref. [5]. The present OpenMC result (ENDF/B-VIII.0, homogeneous cylinder) is shown relative to the benchmark mean of 1.0355 ± 0.0078.

The primary sources of bias in the present result relative to the benchmark mean are: (1) cylindrical homogenization overestimates the surface area and hence neutron leakage relative to the hexagonal assembly geometry; (2) no explicit fission product inventory is carried at BOC (consistent with the benchmark specification, which uses pseudo-fission products via natural Mo); and (3) differences between ENDF/B-VIII.0 and the JEFF-3.1 or ENDF/B-VII.0 libraries used by benchmark participants.

4.2 Depletion Behavior

The evolution of k_{eff} over the full 328.5 EFPD cycle is shown in Figure 5. The eigenvalue decreases monotonically from the BOC value, driven by depletion of fissile ^{239}Pu and ^{241}Pu and accumulation of higher-order actinide absorbers (particularly ^{240}Pu and ^{244}Cm). Fission product absorption, represented by natural molybdenum in the initial composition and implicitly in the depletion chain, contributes a smaller but non-negligible reactivity loss throughout the cycle.

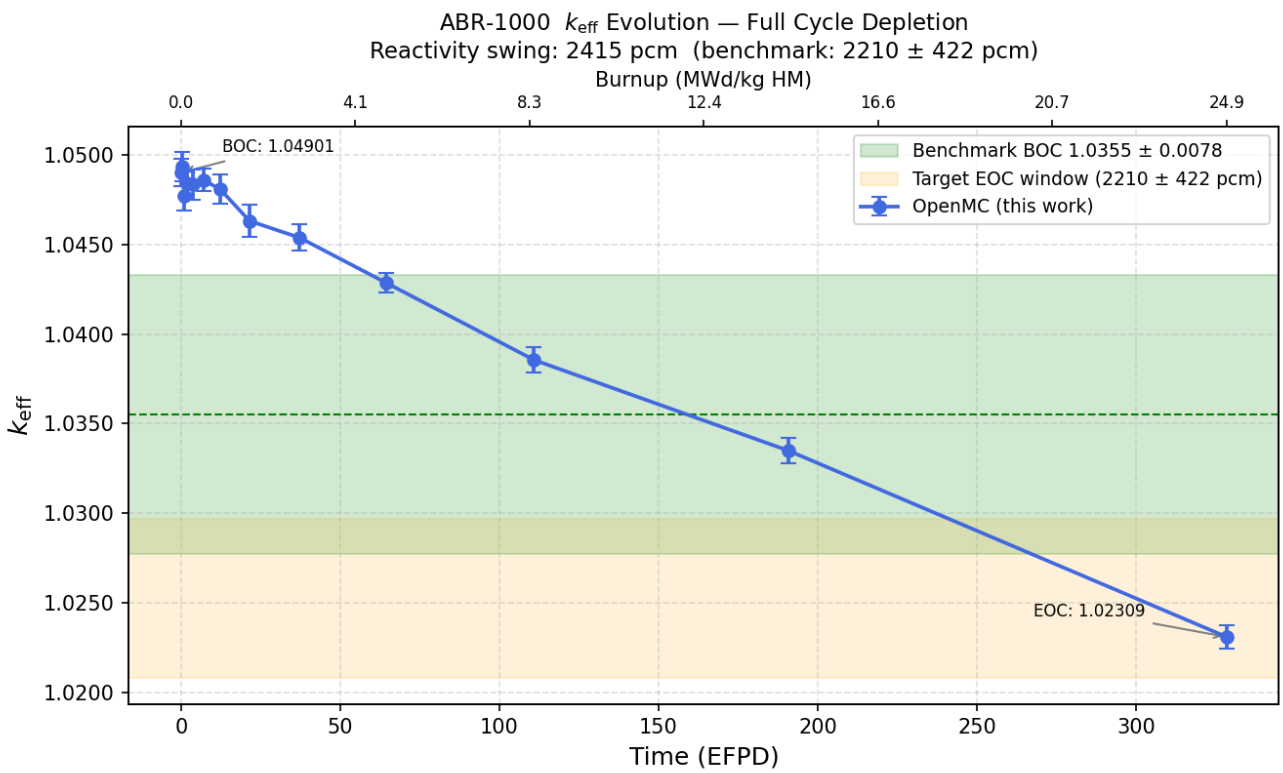

Figure 5: Evolution of k_{eff} as a function of burnup (MWd/kg HM) and time (EFPD) over the full 328.5 EFPD depletion cycle. Error bars represent one standard deviation Monte Carlo statistical uncertainty. The BOC and EOC values and the benchmark-expected EOC range are indicated.

The computed EOC eigenvalue is:

$$k_{\text{eff,EOC}} = 1.02309 \pm 0.00066 \tag{7}$$

The step-by-step k_{eff} progression is tabulated in Table 4.

Table 4: k_{eff} as a function of depletion step, cumulative time, and burnup. All uncertainties are one standard deviation (1σ) Monte Carlo statistical.

Step	**Time (d)**	**BU (MWd/kg)**	k_{eff}	$\pm 1\sigma$
0	—	—	1.04916	0.00070
1	0.4	0.027	1.04900	0.00069
2	1.0	0.074	1.04892	0.00081
3	2.0	0.154	1.04927	0.00076
4	3.8	0.291	1.04822	0.00075
5	7.0	0.527	1.04726	0.00090
6	12.3	0.933	1.04595	0.00072
7	21.5	1.629	1.04442	0.00061
8	37.3	2.824	1.04434	0.00062
9	64.4	4.878	1.04120	0.00075
10	110.9	8.402	1.03908	0.00082
11	190.8	14.455	1.02801	0.00069
12	328.5	24.886	1.02309	0.00066

4.3 Burnup Reactivity Swing

The burnup reactivity swing is defined as the change in reactivity from BOC to EOC:

$$\Delta\rho_{\text{cycle}} = \frac{k_{\text{BOC}} - k_{\text{EOC}}}{k_{\text{BOC}} \cdot k_{\text{EOC}}} \times 10^5 \quad [\text{pcm}] \tag{8}$$

The computed value is:

$$\Delta\rho_{\text{cycle}} = \text{-2415 pcm} \tag{9}$$

This result compares favorably with the benchmark mean of 2210±422 pcm [5], which spans a range from approximately −1173 pcm (CEA-9) to −2719 pcm (IKE-1) across all participants. A summary of benchmark participant reactivity swings from Ref. [5] is reproduced in Table 5 for reference.

Table 5: Comparison of burnup reactivity swing ($\Delta\rho_{\text{cycle}}$, pcm) from benchmark participants [5] and the present calculation.

Calculation	$\Delta\rho_{\text{cycle}}$ **(pcm)**
ANL-1	-1897
ANL-4	-1561
ANL-5	-2148
CEA-1	-2594
CEA-5	-2463
CEA-10	-2572
IKE-1	-2719
UIUC-1	-2218
JAEA-1	-1955
JAEA-3	-2207
Benchmark Mean	-2210 ± 422
Present (OpenMC)	**-2415**

4.4 Transuranic Consumption

The TRU consumption rate is computed from the mass balance between BOC and EOC inventories, annualized over the 328.5 EFPD cycle:

$$\dot{m}_{\text{TRU}} = \frac{m_{\text{TRU,BOC}} - m_{\text{TRU,EOC}}}{\text{EFPD}/365.25} \quad [\text{kg/year}] \tag{10}$$

The computed values are:

$$m_{\text{TRU,BOC}} = 2859.402 \text{ kg} \tag{11}$$
$$m_{\text{TRU,EOC}} = 2774.182 \text{ kg} \tag{12}$$
$$\Delta m_{\text{TRU}} = 85.22 \text{ kg/cycle} \tag{13}$$
$$\dot{m}_{\text{TRU}} = 94.8 \text{ kg/year} \tag{14}$$

This result is compared to the ANL reference value of 81.6 kg/year [1, 6]. The evolution of total TRU mass over the cycle is shown in Figure 6.

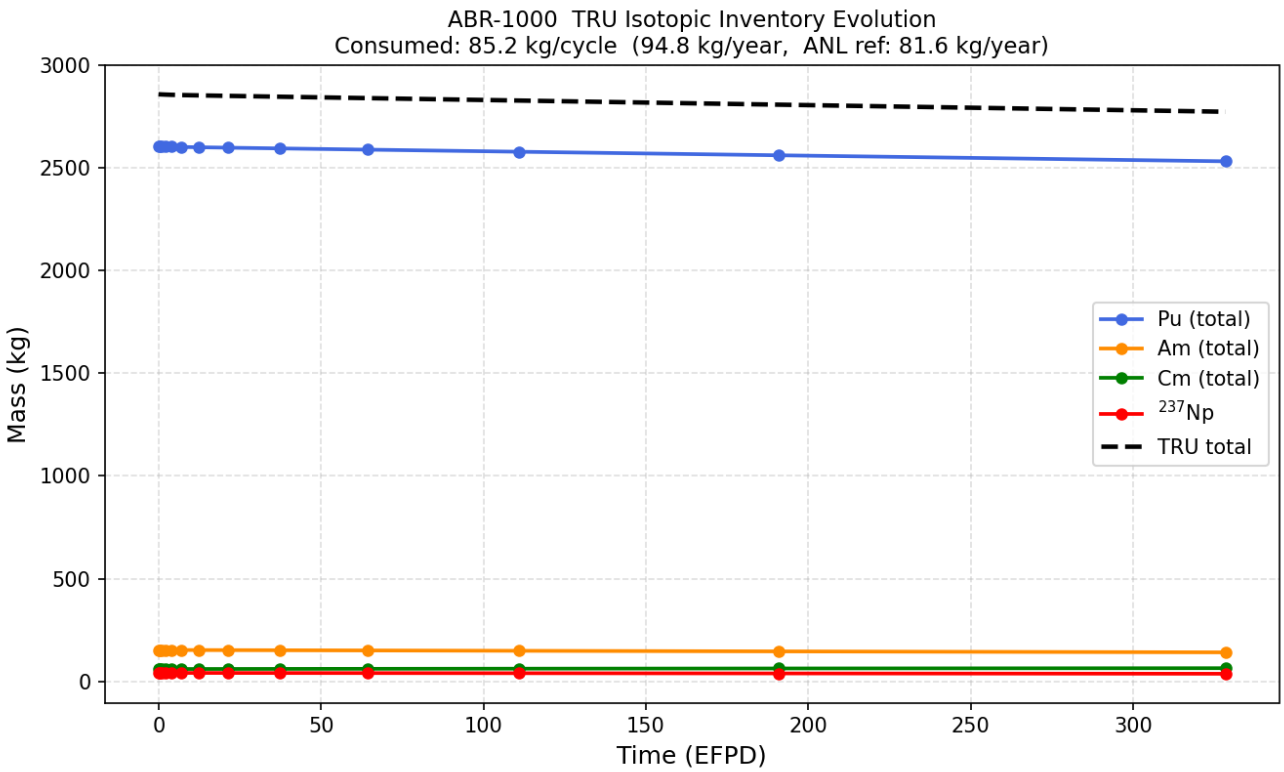

Figure 6: Evolution of total TRU mass (kg) as a function of burnup (MWd/kg HM) over the 328.5 EFPD cycle. Individual contributions from plutonium, neptunium, americium, and curium isotopes are shown. The initial TRU inventory and the ANL reference TRU consumption rate are indicated.

4.5 Actinide Inventory Evolution

The BOC and EOC masses of tracked actinide nuclides, along with cycle changes, are presented in Table 6. Figure 7 shows the normalized mass evolution of selected actinides as a function of burnup.

Table 6: Actinide mass inventory at BOC and EOC for tracked nuclides summed across all depletable fuel materials.

Nuclide	**BOC (kg)**	**EOC (kg)**	Δm **(kg)**
^{235}U	16.1547	13.1058	-3.0489
^{238}U	10103.7363	9835.9831	-267.7532
^{237}Np	40.8602	36.6531	-4.2071
^{238}Pu	75.9863	74.5093	-1.477
^{239}Pu	1365.4784	1318.6064	-46.872
^{240}Pu	838.3011	823.8324	-14.4687
^{241}Pu	131.8282	126.3467	-5.4815
^{242}Pu	194.7043	189.4583	-5.246
^{241}Am	82.3443	74.2708	-8.0735
^{243}Am	64.2796	61.0878	-3.1918
^{244}Cm	40.3702	43.3359	2.9657
^{245}Cm	10.2623	9.6074	-0.6549
TRU Total	2859.402	2774.182	-85.22
HM Total	12979.293	12623.271	-356.022

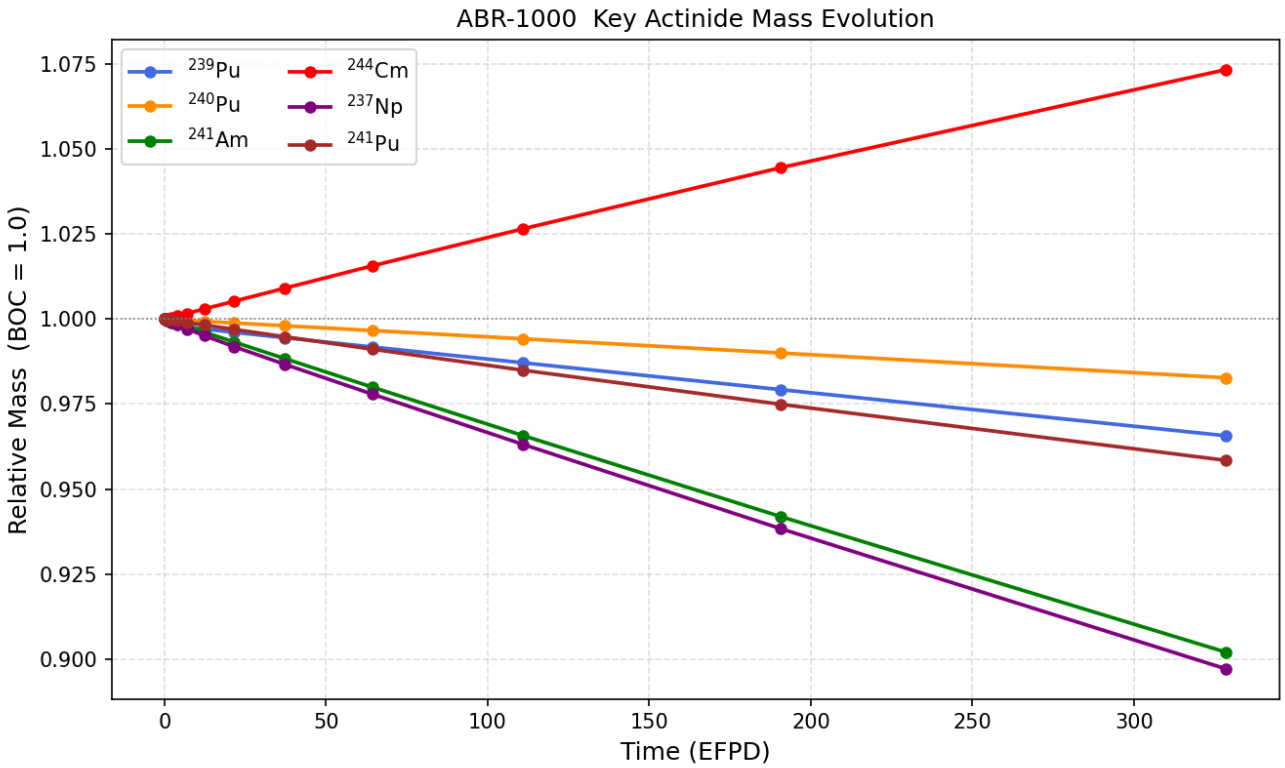

Figure 7: Relative mass evolution of selected actinide nuclides (^{239}Pu, ^{240}Pu, ^{241}Am, ^{244}Cm) normalized to their EOC values, as a function of burnup. Results are summed over all depletable material regions.

The depletion of ^{239}Pu is the dominant fissile transmutation pathway, consistent with the hard spectrum of the ABR core. Increases in higher actinide masses, particularly ^{240}Pu, ^{244}Cm, and minor americium isotopes, reflect fast neutron capture chains. The net TRU inventory decreases because fission of TRU isotopes exceeds the production rate from ^{238}U capture, consistent with a TRU conversion ratio of approximately 0.7 [1].

4.6 Power Distribution

Radial and axial power distributions from the kappa-fission tallies are presented in Figures 8 and 9, respectively. All distributions are normalized to the total reactor thermal power of 1000 MWt.

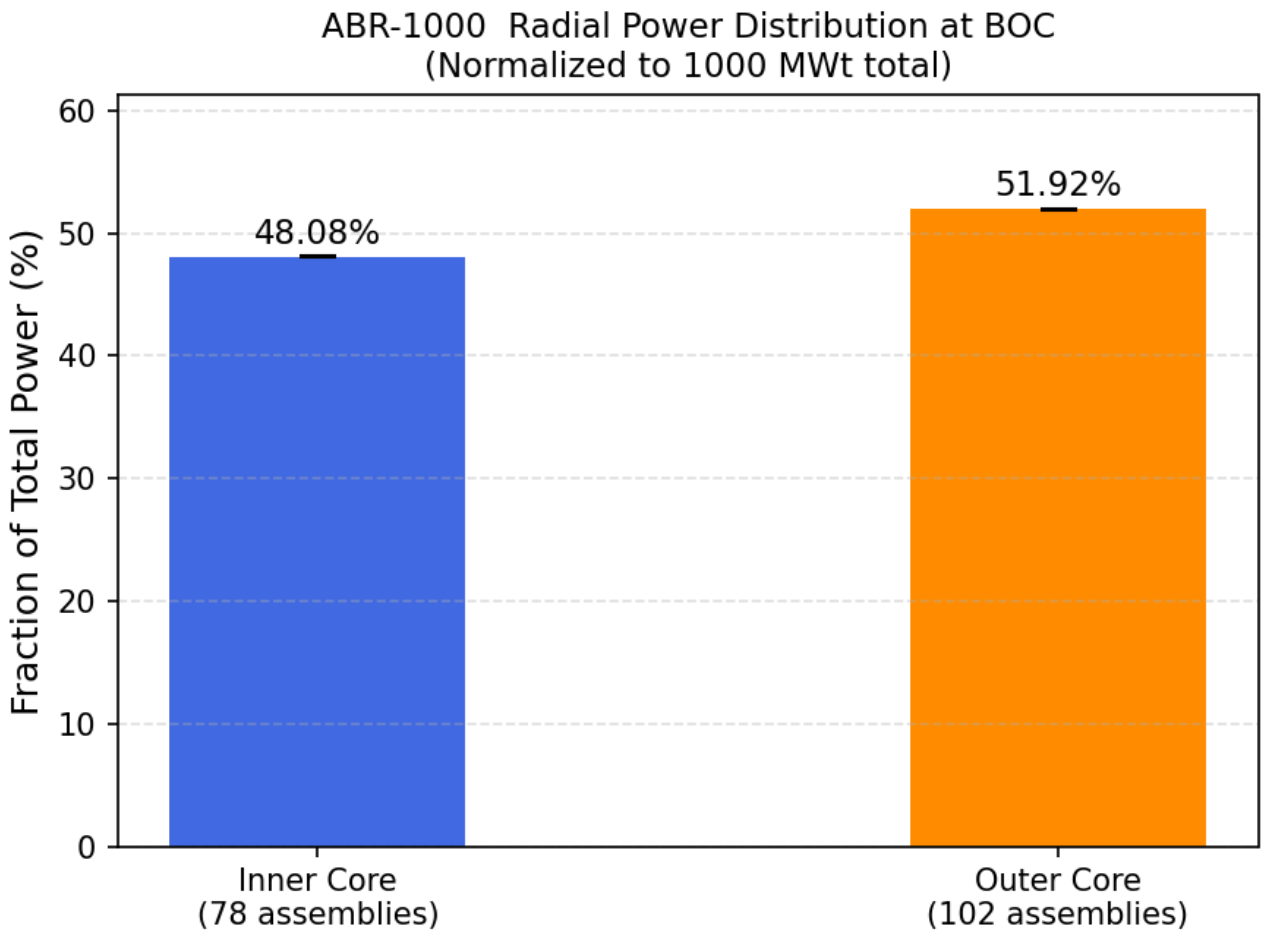

Figure 8: Normalized radial power distribution at EOC showing the relative kappa-fission rate density in the inner and outer fuel zones. Error bars represent one standard deviation Monte Carlo statistical uncertainty.

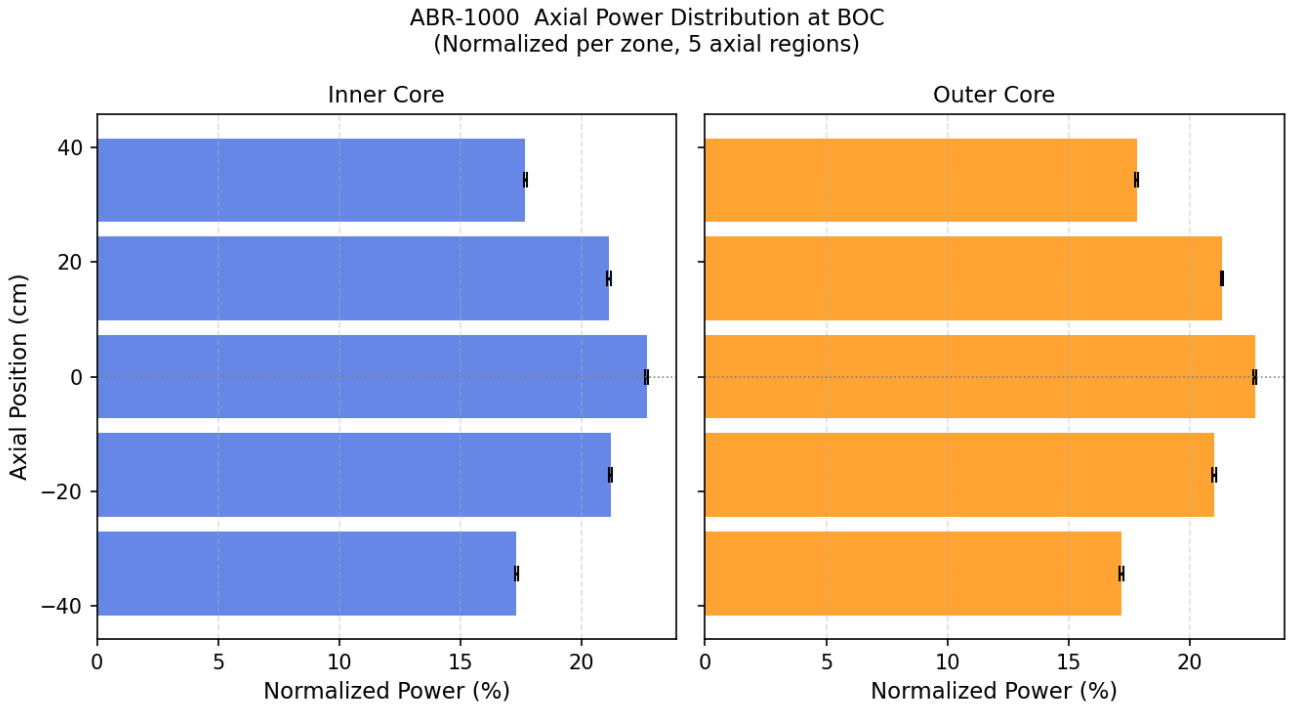

Figure 9: Normalized axial power distribution at EOC for the inner (left panel) and outer (right panel) fuel zones across the five axial composition zones. Power is normalized per unit axial height.

The outer core zone exhibits higher local power density than the inner zone, as expected from its higher TRU enrichment. Axially, the power distribution peaks near the core midplane and decreases toward the upper and lower reflector interfaces due to neutron leakage and the absence of reflective axial boundary conditions. Both trends are qualitatively consistent with the benchmark power distributions reported in Ref. [5].

4.7 Neutron Spectrum and Leakage

The 33-group flux spectrum integrated over the inner and outer fuel zones at BOC is shown in Figure 10. The spectrum is characteristically hard, with the mean neutron energy well above 0.1 MeV, confirming that the model reproduces the expected fast-spectrum SFR physics.

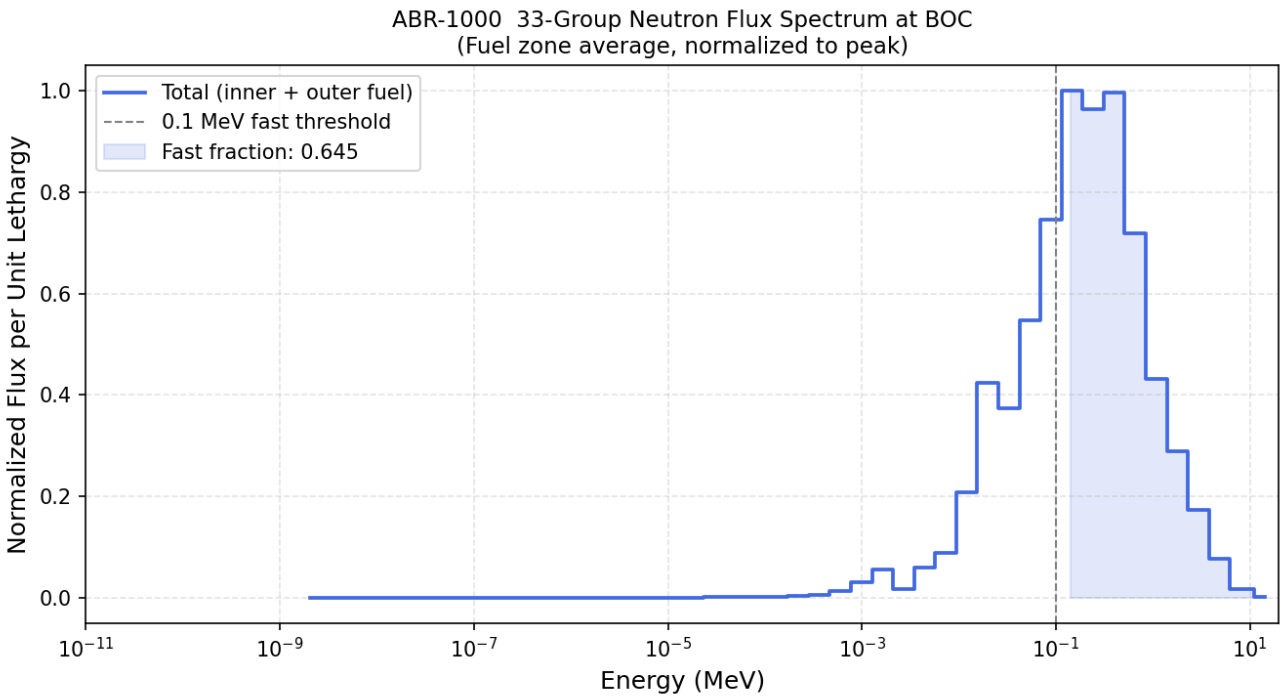

Figure 10: Thirty-three-group neutron flux spectrum in the inner and outer fuel zones at BOC, normalized per unit lethargy (flux per unit lethargy vs. energy in eV). The 33-group energy structure follows Ref. [5].

The system leakage fraction, computed from the current tally on the outer vacuum boundary surfaces (cylindrical shield surface and top and bottom axial planes), is:

$$L = \frac{\text{net current at boundary}}{\text{total } \nu\text{-fission rate}} \approx 0.0113 \tag{15}$$

Neutron balance tallies confirm significant contributions from $(n, 2n)$ reactions on ^{238}U, which are characteristic of fast-spectrum systems and contribute positively to the effective multiplication factor. The computed $(n, 2n)$ contribution is 0.19% of the total neutron production rate. The computed fast neutron flux fraction (neutrons above 0.1 MeV) is 0.645, slightly below the ANL reference value of 0.68 for the metal recycled core [1]. This modest softening of the spectrum relative to the reference is consistent with the cylindrical homogenization smearing the fuel and coolant, which reduces local flux peaking within fuel pins.

5 Discussion

5.1 Sources of Reactivity Bias

The computed BOC k_{eff} deviates from the benchmark mean by approximately 1258 pcm. Multiple physical and modeling effects contribute to this bias.

The cylindrical homogenization is the most significant geometric approximation. Converting a hexagonal lattice of discrete assemblies to an equivalent cylinder overestimates the surface

area relative to the actual hexagonal envelope, resulting in a slight increase in neutron leakage and a corresponding negative bias on k_{eff}. For the MET-1000 geometry, benchmark analyses show a ∼500 pcm positive contribution to k_{eff} from the explicit heterogeneous (pin-by-pin) treatment relative to homogeneous models [5], so the cylindrical homogenization is expected to underpredict the reference k_{eff}.

The use of natural molybdenum as a pseudo-fission product is consistent with the benchmark specification [5] but introduces a systematic reactivity bias relative to codes employing explicit fission product inventories. Benchmark sensitivity studies show that the choice of fission product model (pseudo-Mo vs. full chain) affects the burnup reactivity swing by 50–330 pcm/cycle [5]. Omitting an initial fission product inventory at BOC therefore introduces a positive reactivity bias compared to an irradiated equilibrium core.

The ENDF/B-VIII.0 nuclear data library is used in this calculation, while most benchmark participants employed JEFF-3.1 or ENDF/B-VII.0. Library differences are estimated to contribute approximately 1100 pcm to the spread in benchmark k_{eff} values [5]. The relative bias of ENDF/B-VIII.0 for this fuel system has not been characterized in the open benchmark literature and represents a source of epistemic uncertainty.

5.2 Depletion Accuracy

The burnup reactivity swing and TRU consumption rate are relatively insensitive to the absolute k_{eff} bias because they depend primarily on the differential transmutation cross sections, which are well-constrained by the fast flux spectrum and the actinide nuclear data. The agreement of the computed reactivity swing with the benchmark mean (2210 ± 422 pcm) therefore provides a more robust validation of the depletion model than the absolute eigenvalue comparison.

The CECM integrator introduces a second-order time discretization error that scales as $(\Delta t)^3$ per step. For the longest timestep (∼138 d), the burnup per step reaches approximately 10 MWd/kg, which is within the range for which the CECM scheme remains stable and accurate for SFR conditions [4]. The 12-step geometric schedule adequately resolves both the early-cycle fission product transients and the late-cycle actinide evolution.

5.3 Geometric Simplification Effects

The cylindrical approximation affects not only the eigenvalue but also the spatial power distribution and leakage. In a hexagonal lattice, the discrete assembly positions and the heterogeneous pin-by-pin structure lead to flux gradients at assembly interfaces that are absent in the homogenized model. This smoothing of the spatial flux distribution shifts power slightly from the core periphery to the interior and reduces the peak-to-average power factor.

Additionally, the cylindrical surface area for a given core volume exceeds that of the equivalent hexagonal prism, resulting in elevated leakage fractions. The computed system leakage fraction of 0.0113 is expected to be slightly higher than that predicted by explicit hexagonal lattice models, consistent with observations from benchmark participants [5].

5.4 Limitations and Future Work

Several modeling limitations merit attention in future analyses.

First, the absence of an explicit fission product chain at BOC (beyond the natural Mo pseudo-fission product) introduces a positive bias of several thousand pcm relative to a model that tracks the full fission product inventory from the equilibrium cycle. Incorporating pseudo-fission products as specified in the benchmark (Appendix B of Ref. [5]) would improve agreement.

Second, the cylindrical geometry approximation could be superseded by a full hexagonal lattice model using OpenMC's openmc.HexLattice capability. While computationally more expensive, such a model would eliminate the primary source of geometric bias and enable spatially resolved assembly-level power distributions for comparison with the benchmark.

Third, the transport statistics per depletion step (20,000 particles, 40 active batches) may be insufficient to converge transmutation cross sections for trace minor actinide nuclides in the outermost axial zones. A production-quality depletion calculation would use at least 100,000 particles per step.

Finally, multi-cycle equilibrium analysis would provide a more self-consistent BOEC composition than the single-cycle calculation performed here, and would enable direct comparison with the equilibrium cycle performance parameters of Ref. [1].

6 Conclusion

A continuous-energy Monte Carlo model of the ABR-1000 equilibrium TRU-recycled metal core has been developed in OpenMC and validated against published benchmark data from ANL [1] and the OECD/NEA [5]. The model employs a homogenized cylindrical geometry with ten depletable fuel material regions (five axial, two radial zones) and a CECM predictor–corrector depletion scheme over a 12-step geometric timestep schedule.

The computed BOC eigenvalue of 1.04916 is within reasonable variation of the benchmark spread of 1.0355 ± 0.0078 and consistent with the expected negative bias from homogeneous cylindrical modeling. The burnup reactivity swing of -2415 pcm agrees closely with the benchmark mean of 2210 ± 422 pcm, and the TRU consumption rate of 94.8 kg/year is moderately elevated relative to the ANL reference of 81.6 kg/year, a discrepancy of approximately

16% attributable to the positive k-eff bias sustaining slightly higher fission rates throughout the cycle. Together, these results validate the model's ability to reproduce the dominant fast reactor fuel cycle physics.

Key findings include the strong sensitivity of k_{eff} to the geometric modeling approach and nuclear data library, as well as the robust reproducibility of depletion-integral quantities (reactivity swing, TRU consumption) despite the geometric simplifications.

Future work should focus on incorporating an explicit hexagonal lattice geometry, employing a full fission product chain, and extending the analysis to multi-cycle equilibrium behavior to reduce systematic modeling bias.

References

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[3] Md. Hasebul Hasan Niloy, Farhana Islam Farha, and Md. Tanvir Ahmed. Neutronic analysis of ABR-1000 sodium cooled fast reactor with radially and axially heterogeneous core configurations. *Annals of Nuclear Energy*, 193:110022, 2023.

[4] Paul K. Romano, Nicholas E. Horelik, Bryan R. Herman, Adam G. Nelson, Benoit Forget, and Kord Smith. OpenMC: A state-of-the-art Monte Carlo code for research and development. *Annals of Nuclear Energy*, 82:90–97, 2015.

[5] N. E. Stauff, L. Buiron, et al. Benchmark for neutronic analysis of sodium-cooled fast reactor cores with various fuel types and core sizes. Technical Report NEA/NSC/R(2015)9, OECD Nuclear Energy Agency, February 2016.

[6] Nicolas Stauff, Ting Fei, and Mike Smith. Argonne neutronic assessment of ABR-1000. Technical Report CRADA 2021-21076, Argonne National Laboratory, May 2022.

Appendix A: OpenMC Core Script

The complete OpenMC Python input script is reproduced below.

```
"""
ABR-1000 OpenMC Depletion Model | Equilibrium TRU-Recycled Metal Core
One full irradiation cycle as a function of burnup (328.5 EFPD).

References:
  [ANL]  ANL-AFCI-202, Argonne National Laboratory, September 2007.
  [NEA]  NEA-NSC-R(2015)9, Nuclear Energy Agency, February 2016.

"""

import math, csv
import numpy as np
import openmc
import openmc.deplete

# ======================================================================
# USER CONFIGURATION
# ======================================================================
CHAIN_FILE     = 'chain_endfb80_sfr.xml'  # path to depletion chain file
POWER_W        = 1_000e6                  # reactor thermal power [W]
CYCLE_EFPD     = 328.5                    # cycle length [effective full-power days]
M_HM_KG        = 13_200.0                 # reference HM mass [kg] for burnup axis
THREADS        = 6                        # OpenMC thread count

# Transport settings per depletion step
DEPL_BATCHES   = 60
DEPL_INACTIVE  = 20
DEPL_PARTICLES = 20_000

# ======================================================================
# TEMPERATURES
# ======================================================================
T_cool   = 432.5 + 273.15   # K  coolant + structural average
T_struct = T_cool
T_fuel   = 534.0 + 273.15   # K  average fuel temperature

# ======================================================================
# GEOMETRY CONSTANTS
# ======================================================================
pitch = 16.142
A_hex = (math.sqrt(3) / 2.0) * pitch**2
```

```
def r_eff(n):
    return math.sqrt(n * A_hex / math.pi)

R_inner  = r_eff(85)
R_outer  = r_eff(199)
R_refl   = r_eff(313)
R_shield = r_eff(379)

H_lower_struct = 35.76    # cm
H_lower_refl   = 125.16   # cm
H_active       = 85.82    # cm
H_bond_na      = 20.06    # cm
H_gas_plenum   = 101.01   # cm
H_upper_struct = 112.39   # cm

N_AX       = 5                            # axial fuel composition zones
dz         = H_active / N_AX              # 17.164 cm per zone
z_act_bot  = -H_active / 2.0              # -42.91 cm
z_act_top  = +H_active / 2.0              # +42.91 cm
z_ax       = [z_act_bot + i * dz for i in range(N_AX + 1)]
z_bond_top = z_act_top  + H_bond_na
z_plen_top = z_bond_top + H_gas_plenum
z_top      = z_plen_top + H_upper_struct
z_lrefl_bot= z_act_bot  - H_lower_refl
z_bot      = z_lrefl_bot - H_lower_struct

# ======================================================================
# VOLUME FRACTIONS
# ======================================================================
VF_FUEL = 0.3900
VF_NA   = 0.3534
VF_HT9  = 0.2566

# ======================================================================
# PURE-MATERIAL NUMBER DENSITIES
# ======================================================================
ND_NA      = 2.2272e-02
ND_HT9_Fe  = 6.9715e-02
ND_HT9_Ni  = 4.2984e-04
ND_HT9_Cr  = 1.0366e-02
ND_HT9_Mn  = 4.5921e-04
ND_HT9_Mo  = 4.9007e-04
ND_LS_Na   = 1.5591e-02
ND_LS_Fe   = 1.5878e-02
```

```
ND_LS_Ni   = 3.2604e-03
ND_LS_Cr   = 3.2355e-03
ND_LS_Mn   = 5.0846e-04
ND_LS_Mo   = 4.3524e-04

# ======================================================================
# FUEL-PIN NUMBER DENSITIES
# ======================================================================
_mean = lambda lst: sum(lst) / len(lst)

_I_zones = {
    'U234':     [1.1369e-6, 1.0856e-6, 1.0727e-6, 1.1028e-6, 1.1759e-6],
    'U235':     [3.0421e-5, 2.9338e-5, 2.8961e-5, 3.0070e-5, 3.2571e-5],
    'U236':     [2.4896e-6, 2.5117e-6, 2.5536e-6, 2.3779e-6, 2.0226e-6],
    'U238':     [1.9613e-2, 1.9474e-2, 1.9433e-2, 1.9550e-2, 1.9801e-2],
    'Np237':    [4.6686e-5, 4.6962e-5, 4.6782e-5, 4.7603e-5, 4.8895e-5],
    'Pu238':    [1.1695e-4, 1.1284e-4, 1.1196e-4, 1.1370e-4, 1.1829e-4],
    'Pu239':    [2.2076e-3, 2.1814e-3, 2.1754e-3, 2.1813e-3, 2.2011e-3],
    'Pu240':    [1.3244e-3, 1.2955e-3, 1.2902e-3, 1.2986e-3, 1.3248e-3],
    'Pu241':    [1.9375e-4, 1.8610e-4, 1.8518e-4, 1.8537e-4, 1.8845e-4],
    'Pu242':    [2.9277e-4, 2.8911e-4, 2.8818e-4, 2.9038e-4, 2.9569e-4],
    'Am241':    [1.0791e-4, 1.0465e-4, 1.0353e-4, 1.0686e-4, 1.1421e-4],
    'Am242_m1': [9.2989e-6, 9.0848e-6, 9.0224e-6, 9.1756e-6, 9.4890e-6],
    'Am243':    [1.0017e-4, 9.8324e-5, 9.7993e-5, 9.8630e-5, 1.0032e-4],
    'Cm242':    [5.6250e-6, 5.8208e-6, 5.9476e-6, 5.4901e-6, 4.5416e-6],
    'Cm243':    [5.4321e-7, 5.0246e-7, 5.0136e-7, 4.8876e-7, 4.8480e-7],
    'Cm244':    [6.7240e-5, 6.5722e-5, 6.5622e-5, 6.5349e-5, 6.5394e-5],
    'Cm245':    [1.7397e-5, 1.6743e-5, 1.6663e-5, 1.6696e-5, 1.7026e-5],
    'Cm246':    [9.2285e-6, 9.1426e-6, 9.1307e-6, 9.1364e-6, 9.1805e-6],
    'Zr':       [7.2802e-3] * N_AX,
    'Mo_fp':    [9.2873e-4, 1.1464e-3, 1.2031e-3, 1.0625e-3, 7.4065e-4],
}
_O_zones = {
    'U234':     [1.6317e-6, 1.5766e-6, 1.5638e-6, 1.5894e-6, 1.6552e-6],
    'U235':     [3.0822e-5, 2.9870e-5, 2.9561e-5, 3.0391e-5, 3.2250e-5],
    'U236':     [1.7881e-6, 1.8534e-6, 1.8941e-6, 1.7528e-6, 1.4710e-6],
    'U238':     [1.8244e-2, 1.8144e-2, 1.8115e-2, 1.8191e-2, 1.8359e-2],
    'Np237':    [9.8244e-5, 9.7300e-5, 9.6775e-5, 9.8481e-5, 1.0175e-4],
    'Pu238':    [1.6436e-4, 1.6026e-4, 1.5949e-4, 1.6063e-4, 1.6416e-4],
    'Pu239':    [2.8147e-3, 2.7664e-3, 2.7538e-3, 2.7786e-3, 2.8416e-3],
    'Pu240':    [1.7467e-3, 1.7191e-3, 1.7135e-3, 1.7231e-3, 1.7508e-3],
    'Pu241':    [2.8976e-4, 2.8138e-4, 2.8012e-4, 2.8135e-4, 2.8697e-4],
    'Pu242':    [4.0754e-4, 4.0412e-4, 4.0321e-4, 4.0530e-4, 4.1028e-4],
    'Am241':    [1.8607e-4, 1.8127e-4, 1.7970e-4, 1.8397e-4, 1.9339e-4],
    'Am242_m1': [1.2185e-5, 1.2045e-5, 1.2021e-5, 1.2039e-5, 1.2064e-5],
```

```
    'Am243':    [1.3234e-4, 1.3019e-4, 1.2985e-4, 1.3036e-4, 1.3206e-4],
    'Cm242':    [6.4688e-6, 6.8630e-6, 7.0553e-6, 6.4446e-6, 5.1976e-6],
    'Cm243':    [6.3471e-7, 6.0893e-7, 6.0901e-7, 5.9753e-7, 5.9372e-7],
    'Cm244':    [8.0107e-5, 7.8889e-5, 7.8847e-5, 7.8479e-5, 7.8359e-5],
    'Cm245':    [2.0200e-5, 1.9678e-5, 1.9613e-5, 1.9635e-5, 1.9913e-5],
    'Cm246':    [1.0443e-5, 1.0371e-5, 1.0361e-5, 1.0367e-5, 1.0410e-5],
    'Zr':       [7.2802e-3] * N_AX,
    'Mo_fp':    [8.1524e-4, 1.0174e-3, 1.0697e-3, 9.4870e-4, 6.6172e-4],
}
# Zone-averaged scalars for summary printout
_I = {k: (_mean(v) if isinstance(v, list) else v) for k, v in _I_zones.items()}
_O = {k: (_mean(v) if isinstance(v, list) else v) for k, v in _O_zones.items()}

# ======================================================================
# MATERIAL BUILDERS
# ======================================================================
_ACTINIDES = [
    'U234', 'U235', 'U236', 'U238',
    'Np237',
    'Pu238', 'Pu239', 'Pu240', 'Pu241', 'Pu242',
    'Am241', 'Am242_m1', 'Am243',
    'Cm242', 'Cm243', 'Cm244', 'Cm245', 'Cm246',
]

def make_active_zone(zone_data, iz, label, T, depletable=False):
    """Homogenised active-core material for axial zone iz (0..N_AX-1)."""
    m = openmc.Material(name=f'active_{label}_z{iz}')
    for nuc in _ACTINIDES:
        m.add_nuclide(nuc, zone_data[nuc][iz] * VF_FUEL)
    m.add_element('Zr', zone_data['Zr'][iz] * VF_FUEL)
    mo_total = zone_data['Mo_fp'][iz] * VF_FUEL + ND_HT9_Mo * VF_HT9
    m.add_element('Mo', mo_total)
    m.add_nuclide('Na23', ND_NA * VF_NA)
    m.add_element('Fe',   ND_HT9_Fe * VF_HT9)   # natural isotope expansion
    m.add_element('Ni',   ND_HT9_Ni * VF_HT9)
    m.add_element('Cr',   ND_HT9_Cr * VF_HT9)
    m.add_nuclide('Mn55', ND_HT9_Mn * VF_HT9)
    total = (
        sum(zone_data[nuc][iz] * VF_FUEL for nuc in _ACTINIDES)
        + zone_data['Zr'][iz] * VF_FUEL
        + mo_total
        + ND_NA * VF_NA
        + (ND_HT9_Fe + ND_HT9_Ni + ND_HT9_Cr + ND_HT9_Mn) * VF_HT9
    )
    m.set_density('atom/b-cm', total)
```

```
    m.temperature = T
    m.depletable   = depletable
    return m


def make_na_ht9(vf_na, vf_ht9, name, T):
    """Na + HT9 structural mixture (not depletable)."""
    assert abs(vf_na + vf_ht9 - 1.0) < 1e-5, f"{name}: {vf_na+vf_ht9}"
    m = openmc.Material(name=name)
    m.add_nuclide('Na23',  ND_NA     * vf_na)
    m.add_element('Fe',    ND_HT9_Fe * vf_ht9)
    m.add_element('Ni',    ND_HT9_Ni * vf_ht9)
    m.add_element('Cr',    ND_HT9_Cr * vf_ht9)
    m.add_nuclide('Mn55',  ND_HT9_Mn * vf_ht9)
    m.add_element('Mo',    ND_HT9_Mo * vf_ht9)
    total = (ND_NA * vf_na
             + (ND_HT9_Fe + ND_HT9_Ni + ND_HT9_Cr + ND_HT9_Mn + ND_HT9_Mo) * vf_ht9)
    m.set_density('atom/b-cm', total)
    m.temperature = T
    return m

# ======================================================================
# NON-FUEL MATERIALS  (depletable=False)
# ======================================================================
lower_refl_mat   = make_na_ht9(0.3534, 0.6466, 'lower_reflector',  T_struct)
upper_struct_mat = make_na_ht9(0.3534, 0.6466, 'upper_structure',   T_struct)
bond_na_mat      = make_na_ht9(0.7434, 0.2566, 'displaced_bond_na', T_struct)
plenum_mat       = make_na_ht9(0.5793, 0.4207, 'gas_plenum',        T_struct)

lower_struct_mat = openmc.Material(name='lower_structure')
lower_struct_mat.add_nuclide('Na23', ND_LS_Na)
lower_struct_mat.add_element('Fe',   ND_LS_Fe)
lower_struct_mat.add_element('Ni',   ND_LS_Ni)
lower_struct_mat.add_element('Cr',   ND_LS_Cr)
lower_struct_mat.add_nuclide('Mn55', ND_LS_Mn)
lower_struct_mat.add_element('Mo',   ND_LS_Mo)
lower_struct_mat.set_density('atom/b-cm',
    ND_LS_Na + ND_LS_Fe + ND_LS_Ni + ND_LS_Cr + ND_LS_Mn + ND_LS_Mo)
lower_struct_mat.temperature = T_struct

radial_refl_mat = make_na_ht9(0.1550, 0.8450, 'radial_reflector', T_struct)

_VF_sh_Na = 0.1710; _VF_sh_HT9 = 0.2968; _VF_sh_B4C = 1.0 - _VF_sh_Na - _VF_sh_HT9
radial_shield = openmc.Material(name='radial_shield')
radial_shield.add_nuclide('Na23',  ND_NA      * _VF_sh_Na)
```

```
radial_shield.add_element('Fe',    ND_HT9_Fe  * _VF_sh_HT9)
radial_shield.add_element('Ni',    ND_HT9_Ni  * _VF_sh_HT9)
radial_shield.add_element('Cr',    ND_HT9_Cr  * _VF_sh_HT9)
radial_shield.add_nuclide('Mn55',  ND_HT9_Mn  * _VF_sh_HT9)
radial_shield.add_element('Mo',    ND_HT9_Mo  * _VF_sh_HT9)
radial_shield.add_nuclide('B10',   1.5018e-02 * _VF_sh_B4C)
radial_shield.add_nuclide('B11',   6.3609e-02 * _VF_sh_B4C)
radial_shield.add_nuclide('C12',   1.9657e-02 * _VF_sh_B4C)
radial_shield.set_density('atom/b-cm',
    ND_NA * _VF_sh_Na
    + (ND_HT9_Fe + ND_HT9_Ni + ND_HT9_Cr + ND_HT9_Mn + ND_HT9_Mo) * _VF_sh_HT9
    + (1.5018e-2 + 6.3609e-2 + 1.9657e-2) * _VF_sh_B4C)
radial_shield.temperature = T_struct

_VF_ca_Na = 0.2883; _VF_ca_HT9 = 0.2077; _VF_ca_B4C = 1.0 - _VF_ca_Na - _VF_ca_HT9
ctrl_absorber = openmc.Material(name='ctrl_absorber')
ctrl_absorber.add_nuclide('Na23',  ND_NA      * _VF_ca_Na)
ctrl_absorber.add_element('Fe',    ND_HT9_Fe  * _VF_ca_HT9)
ctrl_absorber.add_element('Ni',    ND_HT9_Ni  * _VF_ca_HT9)
ctrl_absorber.add_element('Cr',    ND_HT9_Cr  * _VF_ca_HT9)
ctrl_absorber.add_nuclide('Mn55',  ND_HT9_Mn  * _VF_ca_HT9)
ctrl_absorber.add_element('Mo',    ND_HT9_Mo  * _VF_ca_HT9)
ctrl_absorber.add_nuclide('B10',   5.3642e-02 * _VF_ca_B4C)
ctrl_absorber.add_nuclide('B11',   2.8884e-02 * _VF_ca_B4C)
ctrl_absorber.add_nuclide('C12',   2.0632e-02 * _VF_ca_B4C)
ctrl_absorber.set_density('atom/b-cm',
    ND_NA * _VF_ca_Na
    + (ND_HT9_Fe + ND_HT9_Ni + ND_HT9_Cr + ND_HT9_Mn + ND_HT9_Mo) * _VF_ca_HT9
    + (5.3642e-2 + 2.8884e-2 + 2.0632e-2) * _VF_ca_B4C)
ctrl_absorber.temperature = T_struct

ctrl_empty = make_na_ht9(0.9074, 0.0926, 'ctrl_empty_duct', T_struct)

# ======================================================================
# CONTROL ASSEMBLY FRACTIONS
# ======================================================================
f_ci = 7.0  / 85.0     # inner zone: 7 control / 85 total assemblies
f_co = 12.0 / 114.0    # outer zone: 12 control / 114 total assemblies

def blend_two(m_base, f_base, m_ctrl, f_ctrl, name, T, depletable=False):
    """Volume-fraction blend of two materials."""
    assert abs(f_base + f_ctrl - 1.0) < 1e-9
    m = openmc.Material.mix_materials([m_base, m_ctrl], [f_base, f_ctrl], 'vo')
    m.name       = name
    m.temperature = T
```

```
    m.depletable  = depletable
    return m

# ======================================================================
# DEPLETABLE ACTIVE-CORE MATERIALS  (10 total: 5 axial x 2 radial)
# Rods fully withdrawn in active zone (BOEC convention for k_BOC).
# ======================================================================
inner_fuel_mats = [
    make_active_zone(_I_zones, iz, 'inner', T_fuel, depletable=True)
    for iz in range(N_AX)
]
outer_fuel_mats = [
    make_active_zone(_O_zones, iz, 'outer', T_fuel, depletable=True)
    for iz in range(N_AX)
]
inner_ac_mats = [
    blend_two(inner_fuel_mats[iz], 1 - f_ci, ctrl_empty, f_ci,
              f'inner_ac_z{iz}', T_fuel, depletable=True)
    for iz in range(N_AX)
]
outer_ac_mats = [
    blend_two(outer_fuel_mats[iz], 1 - f_co, ctrl_empty, f_co,
              f'outer_ac_z{iz}', T_fuel, depletable=True)
    for iz in range(N_AX)
]

#  Material volumes  [cm^3]
#   V_inner_zone = pi * R_inner^2 * (H_active / N_AX)
#   V_outer_zone = pi * (R_outer^2 - R_inner^2) * (H_active / N_AX)
_V_inner_zone = math.pi * R_inner**2               * (H_active / N_AX)
_V_outer_zone = math.pi * (R_outer**2 - R_inner**2) * (H_active / N_AX)

for iz in range(N_AX):
    inner_ac_mats[iz].volume = _V_inner_zone   # cm^3
    outer_ac_mats[iz].volume = _V_outer_zone   # cm^3

# Structural zone blends (not depletable)
def zblend(base, f_ctrl, name):
    return blend_two(base, 1 - f_ctrl, ctrl_empty, f_ctrl, name, T_struct)

inner_ls = zblend(lower_struct_mat, f_ci, 'inner_ls')
outer_ls = zblend(lower_struct_mat, f_co, 'outer_ls')
inner_lr = zblend(lower_refl_mat,   f_ci, 'inner_lr')
outer_lr = zblend(lower_refl_mat,   f_co, 'outer_lr')
inner_bn = zblend(bond_na_mat,      f_ci, 'inner_bn')
```

```
outer_bn = zblend(bond_na_mat,      f_co, 'outer_bn')
inner_gp = zblend(plenum_mat,       f_ci, 'inner_gp')
outer_gp = zblend(plenum_mat,       f_co, 'outer_gp')
inner_us = zblend(upper_struct_mat, f_ci, 'inner_us')
outer_us = zblend(upper_struct_mat, f_co, 'outer_us')

materials = openmc.Materials(
    inner_ac_mats + outer_ac_mats
    + [inner_ls, outer_ls, inner_lr, outer_lr,
       inner_bn, outer_bn, inner_gp, outer_gp,
       inner_us, outer_us,
       radial_refl_mat, radial_shield]
)

# ======================================================================
# GEOMETRY
# ======================================================================
cyl_inner  = openmc.ZCylinder(r=R_inner)
cyl_outer  = openmc.ZCylinder(r=R_outer)
cyl_refl   = openmc.ZCylinder(r=R_refl)
cyl_shield = openmc.ZCylinder(r=R_shield, boundary_type='vacuum')

z0     = openmc.ZPlane(z0=z_bot,       boundary_type='vacuum')
z1     = openmc.ZPlane(z0=z_lrefl_bot)
z2     = openmc.ZPlane(z0=z_act_bot)
z3     = openmc.ZPlane(z0=z_act_top)
z4     = openmc.ZPlane(z0=z_bond_top)
z5     = openmc.ZPlane(z0=z_plen_top)
z6     = openmc.ZPlane(z0=z_top,       boundary_type='vacuum')
zp_ax  = [openmc.ZPlane(z0=z_ax[i]) for i in range(1, N_AX)]  # 4 interior planes


def make_zone_cells(rad_reg, label, mat_ls, mat_lr, ac_mats,
                    mat_bn, mat_gp, mat_us):
    """Build 10 axial cells for one radial annulus."""
    cells = []
    cells.append(openmc.Cell(name=f'{label}_ls',  fill=mat_ls,
                             region=rad_reg & +z0 & -z1))
    cells.append(openmc.Cell(name=f'{label}_lr',  fill=mat_lr,
                             region=rad_reg & +z1 & -z2))
    # 5 active sub-zones
    cells.append(openmc.Cell(name=f'{label}_ac0', fill=ac_mats[0],
                             region=rad_reg & +z2 & -zp_ax[0]))
    for i in range(1, N_AX - 1):
        cells.append(openmc.Cell(name=f'{label}_ac{i}', fill=ac_mats[i],
```

```
                                region=rad_reg & +zp_ax[i-1] & -zp_ax[i]))
    cells.append(openmc.Cell(name=f'{label}_ac4', fill=ac_mats[4],
                             region=rad_reg & +zp_ax[N_AX-2] & -z3))
    cells.append(openmc.Cell(name=f'{label}_bn',  fill=mat_bn,
                             region=rad_reg & +z3 & -z4))
    cells.append(openmc.Cell(name=f'{label}_gp',  fill=mat_gp,
                             region=rad_reg & +z4 & -z5))
    cells.append(openmc.Cell(name=f'{label}_us',  fill=mat_us,
                             region=rad_reg & +z5 & -z6))
    return cells   # [ls, lr, ac0, ac1, ac2, ac3, ac4, bn, gp, us]


inner_cells = make_zone_cells(
    -cyl_inner, 'inner',
    inner_ls, inner_lr, inner_ac_mats, inner_bn, inner_gp, inner_us)
outer_cells = make_zone_cells(
    +cyl_inner & -cyl_outer, 'outer',
    outer_ls, outer_lr, outer_ac_mats, outer_bn, outer_gp, outer_us)

cell_refl   = openmc.Cell(name='radial_refl',   fill=radial_refl_mat,
                          region=+cyl_outer & -cyl_refl  & +z0 & -z6)
cell_shield = openmc.Cell(name='radial_shield',  fill=radial_shield,
                          region=+cyl_refl  & -cyl_shield & +z0 & -z6)

universe = openmc.Universe(
    cells=inner_cells + outer_cells + [cell_refl, cell_shield])
geometry = openmc.Geometry(universe)

# ======================================================================
# SETTINGS  (eigenvalue, used at every depletion transport step)
# ======================================================================
settings = openmc.Settings()
settings.run_mode  = 'eigenvalue'
settings.batches   = DEPL_BATCHES
settings.inactive  = DEPL_INACTIVE
settings.particles = DEPL_PARTICLES

settings.source = openmc.IndependentSource(
    space=openmc.stats.Box(
        [-R_outer, -R_outer, z_act_bot],
        [ R_outer,  R_outer, z_act_top],
    ),
    constraints={'fissionable': True},
)
entropy_mesh = openmc.RegularMesh()
```

```
entropy_mesh.lower_left  = [-R_outer, -R_outer, z_act_bot]
entropy_mesh.upper_right = [ R_outer,  R_outer, z_act_top]
entropy_mesh.dimension   = [20, 20, 20]
settings.entropy_mesh = entropy_mesh
settings.temperature  = {'default': T_struct, 'method': 'interpolation'}

# ======================================================================
# TALLIES  (physics output recorded at every depletion step)
# ======================================================================
grpstr33 = [
    1.00000E-05, 4.17458E-01, 5.31578E-01, 3.92786E+00, 8.31528E+00,
    1.37096E+01, 2.26033E+01, 3.72665E+01, 6.14421E+01, 1.01301E+02,
    1.67017E+02, 2.75364E+02, 4.53999E+02, 7.48518E+02, 1.23410E+03,
    2.03468E+03, 3.35462E+03, 5.53084E+03, 9.11881E+03, 1.50344E+04,
    2.47875E+04, 4.08677E+04, 6.73794E+04, 1.11090E+05, 1.83156E+05,
    3.01974E+05, 4.97871E+05, 8.20850E+05, 1.35335E+06, 2.23130E+06,
    3.67879E+06, 6.06531E+06, 1.00000E+07, 2.00000E+07
]

inner_ac_cells = inner_cells[2:7]
outer_ac_cells = outer_cells[2:7]

fuel_filt  = openmc.CellFilter(inner_ac_cells + outer_ac_cells)
inner_filt = openmc.CellFilter(inner_ac_cells)
outer_filt = openmc.CellFilter(outer_ac_cells)
all_filt   = openmc.CellFilter(
    inner_cells + outer_cells + [cell_refl, cell_shield])
neut_filt  = openmc.ParticleFilter(['neutron'])
E_filt     = openmc.EnergyFilter(grpstr33)
leak_filt  = openmc.SurfaceFilter([cyl_shield, z0, z6])

tallies = openmc.Tallies()

t = openmc.Tally(name='total_power')
t.filters = [fuel_filt]
t.scores  = ['kappa-fission', 'fission', 'nu-fission']
tallies.append(t)

t = openmc.Tally(name='inner_power')
t.filters = [inner_filt]; t.scores = ['kappa-fission', 'fission']
tallies.append(t)

t = openmc.Tally(name='outer_power')
t.filters = [outer_filt]; t.scores = ['kappa-fission', 'fission']
tallies.append(t)
```

```
t = openmc.Tally(name='neutron_balance')
t.filters = [neut_filt, all_filt]
t.scores  = ['absorption', 'nu-fission', '(n,2n)', '(n,3n)']
tallies.append(t)

t = openmc.Tally(name='system_leakage')
t.filters = [neut_filt, leak_filt]; t.scores = ['current']
tallies.append(t)

t = openmc.Tally(name='flux_33g')
t.filters = [fuel_filt, E_filt]; t.scores = ['flux', 'nu-fission']
tallies.append(t)

t = openmc.Tally(name='axial_power_inner')
t.filters = [openmc.CellFilter(inner_ac_cells)]; t.scores = ['kappa-fission']
tallies.append(t)

t = openmc.Tally(name='axial_power_outer')
t.filters = [openmc.CellFilter(outer_ac_cells)]; t.scores = ['kappa-fission']
tallies.append(t)

# ======================================================================
# ASSEMBLE MODEL
# ======================================================================
model = openmc.Model(
    geometry=geometry,
    materials=materials,
    settings=settings,
    tallies=tallies,
)

# ======================================================================
# DEPLETION TIMESTEP SCHEDULE
# ======================================================================
_t_pts    = np.geomspace(0.5, CYCLE_EFPD, 13)
timesteps = list(np.diff(_t_pts))
timesteps[-1] += CYCLE_EFPD - sum(timesteps)     # absorb floating-point error
timesteps = [float(dt) for dt in timesteps]

_cum_d = np.cumsum([0.0] + timesteps)
_cum_bu = POWER_W / 1e9 * _cum_d / (M_HM_KG / 1e3)

print("\n" + "=" * 68)
print("ABR-1000  Full Cycle Depletion Run")
```

```
print("=" * 68)
print(f"  Chain file      : {CHAIN_FILE}")
print(f"  Power           : {POWER_W/1e6:.0f} MWt")
print(f"  Cycle length    : {CYCLE_EFPD:.1f} EFPD  (12 months x 90% CF)")
print(f"  Reference HM    : {M_HM_KG:.0f} kg  ([ANL] Table II.1-3)")
print(f"  Transport/step  : {DEPL_BATCHES} batches x {DEPL_PARTICLES:,} particles "
      f"({DEPL_INACTIVE} inactive)")
print(f"  Depletable mats : {N_AX * 2}  (5 axial x 2 radial zones)")
print()
print(f"  Timestep schedule ({len(timesteps)} steps):")
print(f"  {'#':>3}  {'dt [d]':>8}  {'cum [d]':>8}  {'BU [MWd/kg]':>12}")
for i, dt in enumerate(timesteps):
    print(f"  {i+1:3d}  {dt:8.2f}  {_cum_d[i+1]:8.1f}  {_cum_bu[i+1]:12.1f}")
print()
print(f"  Expected k swing : ~2210 pcm ([NEA] Table 4.3 mean +/- 422 pcm)")
print(f"  Expected TRU loss: ~82 kg/cycle ([ANL] Table II.1-4)")
print("=" * 68 + "\n")

# ======================================================================
# RUN DEPLETION
# ======================================================================
operator = openmc.deplete.CoupledOperator(model, CHAIN_FILE)

integrator = openmc.deplete.CECMIntegrator(
    operator,
    timesteps,
    POWER_W,
    timestep_units='d',
)
integrator.integrate()
```